where $S := \sum_{j=1}^{n} \lambda_j$ is the sum of the coefficients. Finally, for $\boldsymbol{r}$ to be within $\mathcal{S}$ it has to hold that for any columns m, m' the entries are equal:

$$\begin{aligned} &\lambda_m \alpha + (S - \lambda_m)\beta = \lambda_{m'} \alpha + (S - \lambda_{m'})\beta \\ \Rightarrow \quad &(\lambda_m - \lambda_{m'})(\alpha - \beta) = 0. \end{aligned} \tag{86}$$

But by assumption $\alpha \neq \beta$, and therefore $\lambda_m = \lambda_{m'}$ for all $m, m' \in [n]$. Clearly, if $\lambda_m \neq 0$, then this results in the constant direction produced by any concept block, which is the only direction that lies in $\mathcal{S}$. Therefore each concept block produces $n-1$ linearly independent vectors.

Combined with the redundancy within a concept block, this implies that $\text{rank}(\boldsymbol{Y}) = 1 + k(n-1)$.

Finally, since $\boldsymbol{Y} = \boldsymbol{P}\boldsymbol{X}$,

$$1 + k(n-1) = \text{rank}(\boldsymbol{Y}) \leq \text{rank}(\boldsymbol{P}) \leq d. \tag{87}$$

This proves (75) and shows that such a model requires at least $1 + k(n-1)$ dimensions.

Tightness follows from an explicit construction by placing probes and representations in, for example, orthogonal directions. □

Below, we provide a numerical example to illustrate the form of the logit matrix Y for the case of two concepts, three values each.

Example 1 (Two concepts, three values each: $c = 2,\ n = 3$)**.** Set $(\alpha, \beta) = (1, 0.2)$. The row indices are $(i, j) \in \{1, 2\} \times \{1, 2, 3\}$, the column indices are the $3^2 = 9$ tuples $(v_1, v_2) \in \{1, 2, 3\}^2$:

$$\boldsymbol{Y} = \begin{array}{c} \\ (1,1) \\ (1,2) \\ (1,3) \\ (2,1) \\ (2,2) \\ (2,3) \end{array} \begin{array}{c} \begin{array}{ccccccccc} 11 & 12 & 13 & 21 & 22 & 23 & 31 & 32 & 33 \end{array} \\ \begin{pmatrix} 1 & 1 & 1 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 1 & 1 & 1 & 0.2 & 0.2 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 1 & 1 & 1 \\ 1 & 0.2 & 0.2 & 1 & 0.2 & 0.2 & 1 & 0.2 & 0.2 \\ 0.2 & 1 & 0.2 & 0.2 & 1 & 0.2 & 0.2 & 1 & 0.2 \\ 0.2 & 0.2 & 1 & 0.2 & 0.2 & 1 & 0.2 & 0.2 & 1 \end{pmatrix} \end{array} \tag{88}$$

Note that each block corresponding to either the first or the second concept sum up to the same vector. Within each block there are 2 linearly independent vectors. Hence, $\text{rank}(Y) = 5 = 1 + c(n-1)$.

Under such a design, linear factorization holds immediately – at least up to projection onto the span of the probe vectors; or exactly if the dimensionality of the embeddings is exactly $1 + k(n-1)$. To make the setup explicit, we can also impose a fixed relation between α and β. We avoid the condition of the global mean being zero, as this implies that $\alpha + (n-1)\beta = 0$, i.e., $\alpha = -(n-1)\beta$ (similar to Proposition 6). Related normalization choices are discussed in (Lee et al., 2024).

Proposition 7 (Additive factorization from the on-off pattern)**.** Assume $k \geq 2$ and $n \geq 2$. For each $i \in [k]$ and $j \in [n]$, let $\boldsymbol{w}_{i,j} \in \mathbb{R}^d$ be a probe vector. For each tuple $\boldsymbol{c} = (c_1, \ldots, c_k) \in [n]^k$, let $\boldsymbol{z}_{\boldsymbol{c}} \in \mathbb{R}^d$ be its corresponding representation. Assume there exist $\alpha > \beta$, with $\alpha, \beta \in \mathbb{R}$, such that

$$\boldsymbol{w}_{i,j}^{\intercal} \boldsymbol{z}_{\boldsymbol{c}} = \begin{cases} \alpha & \text{if } j = c_i, \\ \beta & \text{if } j \neq c_i, \end{cases} \qquad \forall i, j, \boldsymbol{c}. \tag{89}$$

Then there exist vectors $\bar{\boldsymbol{z}} \in \mathbb{R}^d$, $\boldsymbol{u}_{i,j} \in \mathbb{R}^d$, and reconstructed representations $\widetilde{\boldsymbol{z}}_{\boldsymbol{c}} \in \mathbb{R}^d$ of the additive form

$$\widetilde{\boldsymbol{z}}_{\boldsymbol{c}} = \bar{\boldsymbol{z}} + \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i}, \qquad \forall \boldsymbol{c} \in [n]^k, \tag{90}$$

such that for every $i, j, \boldsymbol{c}$,

$$\boldsymbol{w}_{i,j}^{\intercal} \widetilde{\boldsymbol{z}}_{\boldsymbol{c}} = \boldsymbol{w}_{i,j}^{\intercal} \boldsymbol{z}_{\boldsymbol{c}} = \begin{cases} \alpha & \text{if } j = c_i, \\ \beta & \text{if } j \neq c_i. \end{cases} \tag{91}$$

So $\widetilde{\boldsymbol{z}}_{\boldsymbol{c}}$ and $\boldsymbol{z}_{\boldsymbol{c}}$ are indistinguishable to the probes.

Proof. We again show that computing the factors as averages satisfies the condition. For that, we define the global mean $\bar{\boldsymbol{z}}$, the concept factors $\boldsymbol{a}_{i,j}$, and the centered factors $\boldsymbol{u}_{i,j}$:

$$\bar{\boldsymbol{z}} := \frac{1}{n^k} \sum_{\boldsymbol{c}' \in [n]^k} \boldsymbol{z}_{\boldsymbol{c}'}, \qquad \boldsymbol{a}_{i,j} := \frac{1}{n^{k-1}} \sum_{\boldsymbol{c}' : c'_i = j} \boldsymbol{z}_{\boldsymbol{c}'}, \qquad \boldsymbol{u}_{i,j} := \boldsymbol{a}_{i,j} - \bar{\boldsymbol{z}}, \qquad \widetilde{\boldsymbol{z}}_{\boldsymbol{c}} := \bar{\boldsymbol{z}} + \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i}.$$

We will confirm this choice works through a simple calculation. First, for any (i, j) the dot product with the average representation is

$$\boldsymbol{w}_{i,j}^{\intercal}\bar{\boldsymbol{z}} = \frac{1}{n^k}\Big(n^{k-1}\alpha + (n^k - n^{k-1})\beta\Big) = \frac{\alpha + (n-1)\beta}{n} =: \delta, \tag{92}$$

Second, for any i', r, we expand the dot product with the average non-centered factor $\boldsymbol{a}_{i,j}$:

$$\boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{a}_{i,j} = \frac{1}{n^{k-1}} \sum_{\boldsymbol{c}':c'_i=j} \boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{z}_{\boldsymbol{c}'}. \tag{93}$$

and decompose it into two cases:

Case 1: $i' = i$ **(probe and conditioning on the same concept).** Here the condition $c'_i = j$ fixes whether the target concept is probed:

- If $r = j$, every term in the sum equals α, so $\boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{a}_{i,j} = \alpha$.
- If $r \neq j$, every term in the sum equals β, so $\boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{a}_{i,j} = \beta$.

Case 2: $i' \neq i$ **(probe and conditioning on different concepts).** Now $c'_i = j$ does not constrain $c'_{i'}$, so:

- exactly n^{k-2} tuples satisfy $c'_{i'} = r$ and contribute α,
- exactly $(n-1)n^{k-2}$ tuples satisfy $c'_{i'} \neq r$ and contribute β.

Therefore

$$\boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{a}_{i,j} = \frac{n^{k-2}\alpha + (n-1)n^{k-2}\beta}{n^{k-1}} = \delta.$$

Putting it all together:

$$\boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{a}_{i,j} = \begin{cases} \alpha, & i' = i,\ r = j, \\ \beta, & i' = i,\ r \neq j, \\ \delta, & i' \neq i. \end{cases}$$

By linearity, subtracting the global mean term gives

$$\boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{u}_{i,j} = \boldsymbol{w}_{i',r}^{\intercal}\boldsymbol{a}_{i,j} - \boldsymbol{w}_{i',r}^{\intercal}\bar{\boldsymbol{z}} = \begin{cases} \alpha - \delta, & i' = i,\ r = j, \\ \beta - \delta, & i' = i,\ r \neq j, \\ 0, & i' \neq i. \end{cases} \tag{94}$$

We now evaluate the reconstructed representation:

$$\begin{aligned} \boldsymbol{w}_{i,j}^{\intercal}\widetilde{\boldsymbol{z}}_{\boldsymbol{c}} &= \boldsymbol{w}_{i,j}^{\intercal}\bar{\boldsymbol{z}} + \sum_{t=1}^{k} \boldsymbol{w}_{i,j}^{\intercal}\boldsymbol{u}_{t,c_t} \\ &= \delta + \boldsymbol{w}_{i,j}^{\intercal}\boldsymbol{u}_{i,c_i} + \sum_{t\neq i} \underbrace{\boldsymbol{w}_{i,j}^{\intercal}\boldsymbol{u}_{t,c_t}}_{=0 \text{ by } (94)} \\ &= \delta + \big(\boldsymbol{w}_{i,j}^{\intercal}\boldsymbol{a}_{i,c_i} - \delta\big) = \boldsymbol{w}_{i,j}^{\intercal}\boldsymbol{a}_{i,c_i} \\ &= \begin{cases} \alpha, & j = c_i, \\ \beta, & j \neq c_i. \end{cases} \end{aligned}$$

Hence $\widetilde{\boldsymbol{z}}_{\boldsymbol{c}}$ reproduces exactly the same on-off probe responses as $\boldsymbol{z}_{\boldsymbol{c}}$ for every concept and value, which proves (91). □

G. Discussion on stability

Here we discuss the role of Stability and what can go wrong without it. Without Stability (Desideratum 3), the only requirement on the readout is that it correctly classifies the observed grid. This places no constraint on how the decision regions are shaped: they can vary arbitrarily across training supports, as long as every grid point "lands" in the correct region. In practice, this means that decision boundaries may pass arbitrarily close to data points, that some concept-value regions may be infinitesimally thin, and that retraining on a different valid support can produce a completely different partition of the space. In short, no robustness holds. As we show below, this pathology does not disappear with scale: even as $n \to \infty$, the decision regions can remain arbitrarily complex, which is never a desirable property. Linear separability alone therefore does not force linear factorization, nor does it force projected R^2 to be close to 1 (see also Sections H.2 and H.4).

Stability prevents these pathologies by requiring that the predicted probability vector over concept values agrees at every input, regardless of which valid support was used for training. In the linear-softmax setting, this implies that probe weights are the same across supports (or, up to a shift in non-binary case), pinning down the decision boundaries and forcing the representational structure established in our main results.

Counterexamples to linear factorization even as $n \to \infty$. Suppose we only assume linear separability on the full grid: for concept tuples $\boldsymbol{q} = (q_1, \ldots, q_k) \in [n]^k$ with embeddings $\boldsymbol{z}_{\boldsymbol{q}} \in \mathbb{R}^d$, there exist probes $\{(\boldsymbol{w}_{i,j}, b_{i,j})\}_{j=1}^n$ such that

$$\arg\max_{j \in [n]} \left(\boldsymbol{w}_{i,j}^\intercal \boldsymbol{z}_{\boldsymbol{q}} + b_{i,j}\right) = q_i, \qquad \forall i \in [k],\ \forall \boldsymbol{q} \in [n]^k. \tag{95}$$

This guarantees perfect classification but does not enforce Stability across supports.

Does this imply additive structure? In general, no. Even in the simple case of $d = 2$, $k = 2$, and even as $n \to \infty$, one can construct point clouds $\{\boldsymbol{z}_{q_1,q_2}\} \subset \mathbb{R}^2$ with perfect linear separability that do *not* admit a decomposition

$$\boldsymbol{z}_{q_1,q_2} = \boldsymbol{u}_{1,q_1} + \boldsymbol{u}_{2,q_2}. \tag{96}$$

Linear separability is therefore compatible with non-factorized geometry. Fig. 18 illustrates this failure mode, which is consistent with the low-R^2 counterexamples in Section H.4.

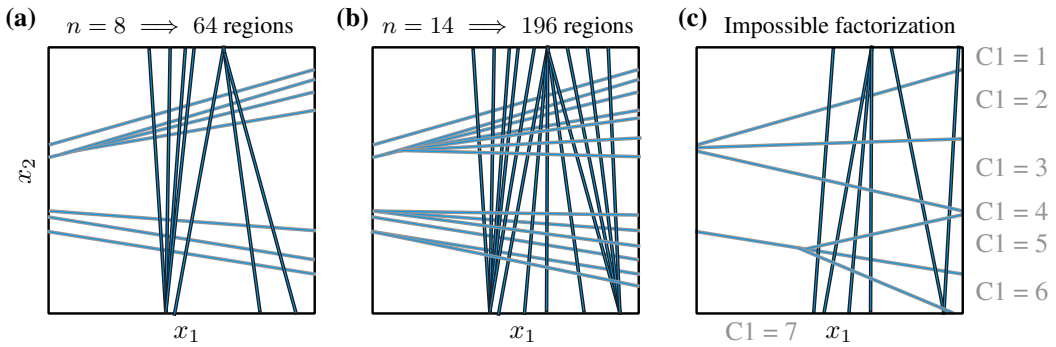

Figure 18. **Linear separability without linear factorization.** Two families of affine decision boundaries in $\mathbb{R}^2$ (black for concept 1, gray for concept 2) divide the plane into regions, one per pair of concept values. **(a,b)**: with $n = 8$ and $n = 14$ levels per concept, the arrangement yields n^2 regions (64 and 196). By inserting additional nearly-parallel boundaries, existing regions can be split into arbitrarily thin pieces while maintaining perfect linear separability. **(c)**: No linear factorization can be achieved: whichever factors we pick, the separability of some datapoints is violated.

Concretely, in Fig. 18, panels **(a)** and **(b)** show two non-parallel families of decision boundaries whose intersections produce n^2 convex cells, one per pair (q_1, q_2). Since perfect classification only requires each grid point to fall in the correct cell, the geometry of the cells themselves is unconstrained. By adding ε-perturbed boundaries, one can subdivide regions so that some concept values occupy arbitrarily small areas as n grows, while perfect separability is preserved. Without Stability, there is nothing to prevent such degenerate configurations, and linear readout constraints alone do not pin down factorized structure. Handling such cases, e.g., by imposing minimum-area constraints on decision regions, is possible in principle but becomes technical; our framework avoids these pathologies by design through the Stability desideratum.

More broadly, any model aiming for robust compositional generalization will need to prevent such degenerate configurations, whether through exact Stability or approximate forms of it.